Part II: Forward Curves & Multi-Commodity Modeling

Commodities Markets and Models

ESILV

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Outline

1. The Dynamics of Forward Curves
   - The Theory of Storage and Convenience Yield
   - Keynes' Theory of Normal Backwardation and Risk Premia
   - The Samuelson Effect

2. Factor Models for Forward Curves
   - Heath-Jarrow-Morton (HJM) Framework for Commodities
   - Non-parametric Approach: PCA of Forward Curves
   - Parametric Models: Long-Term/Short-Term Model

Theory of Storage: Key Concepts

Cash-and-Carry Relationship

For storable commodities, under no-arbitrage:

$$F(t,T) \leq \big(S_t + C(t,T)\big)\big(1 + r(t,T)(T-t)\big)$$

where

- S_t = spot price at t
- $C(t,T)$ = storage cost (discounted to t)
- $r(t,T)$ = interest rate over $[t,T]$
- $F(t,T)$ = forward price for delivery at T

Interpretation:
The forward price $F(t,T)$ **cannot** exceed the cost of buying the commodity today (S_t), storing it ($C(t,T)$), and financing the purchase with borrowed money at the interest rate $r(t,T)$.
If $F(t,T)$ were higher than the right-hand side, arbitrageurs could profit by:

- Selling the forward contract (locking in the higher forward price),
- Borrowing money to buy the commodity at the spot price,
- Storing the commodity until T, and
- Delivering it at T to fulfill the forward contract.

Theory of Storage: Key Concepts (cont.)

Why $F(t, T) <$ RHS?

The inequality $F(t, T) \leq$ RHS can be strict because of the **convenience yield**. Physical holders often prioritize immediate availability over arbitrage profits:

- **Manufacturers** avoid production disruptions by holding inventory.
- **Retailers** ensure supply continuity for clients.
- **Speculators** anticipate scarcity-driven price spikes and prefer holding the commodity.

The **convenience yield** represents the non-monetary benefit of holding the physical commodity, compensating for the lower forward price.

Pedagogical Takeaway

- The Cash-and-Carry Relationship sets an upper bound for forward prices based on spot prices, storage costs, and interest rates.
- The inequality can be strict due to the convenience yield, reflecting the value of immediate availability.
- This framework explains why forward prices may deviate from theoretical no-arbitrage bounds in real-world commodity markets.

Convenience Yield: Definition and Role

Convenience Yield

The **net convenience yield** $\delta(t, T)$ represents the benefit from holding the physical commodity rather than a forward contract, net of storage costs. It reflects the optionality value of physical possession.

Mathematical Formulation

$$F(t, T) = S_t \exp\big((r(t, T) - \delta(t, T))(T - t)\big) \quad \text{(continuous compounding)}$$

or,

$$F(t, T) = \big(S_t - \Delta(t, T)\big)\big(1 + r(t, T)(T - t)\big), \quad \text{(discrete compounding)}$$

Notation Summary

- $\delta(t, T)$ = net convenience yield (annualized)
- $\Delta(t, T) = D(t, T) - C(t, T)$ = net dividend
- $D(t, T)$ = present value of holding benefits
- $C(t, T)$ = present value of storage costs

Convenience Yield: Economic Properties

Key Economic Insights (Kaldor 1939, Working 1949):

- **Negative correlation with inventory**: low inventory $\Rightarrow$ high convenience yield
- High convenience yield $\delta > r \Rightarrow$ **backwardation** (downward-sloping forward curve)
- Low convenience yield $\delta < r \Rightarrow$ **contango** (upward-sloping forward curve)

Pedagogical Takeaway

Three Critical Points:

1. Convenience yield is *implicit*, not directly traded—extracted from forward curves via cost-of-carry arbitrage.
2. It links physical market conditions (inventory levels) to financial prices.
3. Unlike stock dividends (observable), commodity convenience yields are subjective and market-determined.

Theory of Storage: Graphical Evidence

Empirical validation:

- Clear inverse relationship visible in data
- Convenience yield spikes when inventory falls below seasonal norms
- Provides quantitative support for theory of storage

Wheat Forward curve in *contango*

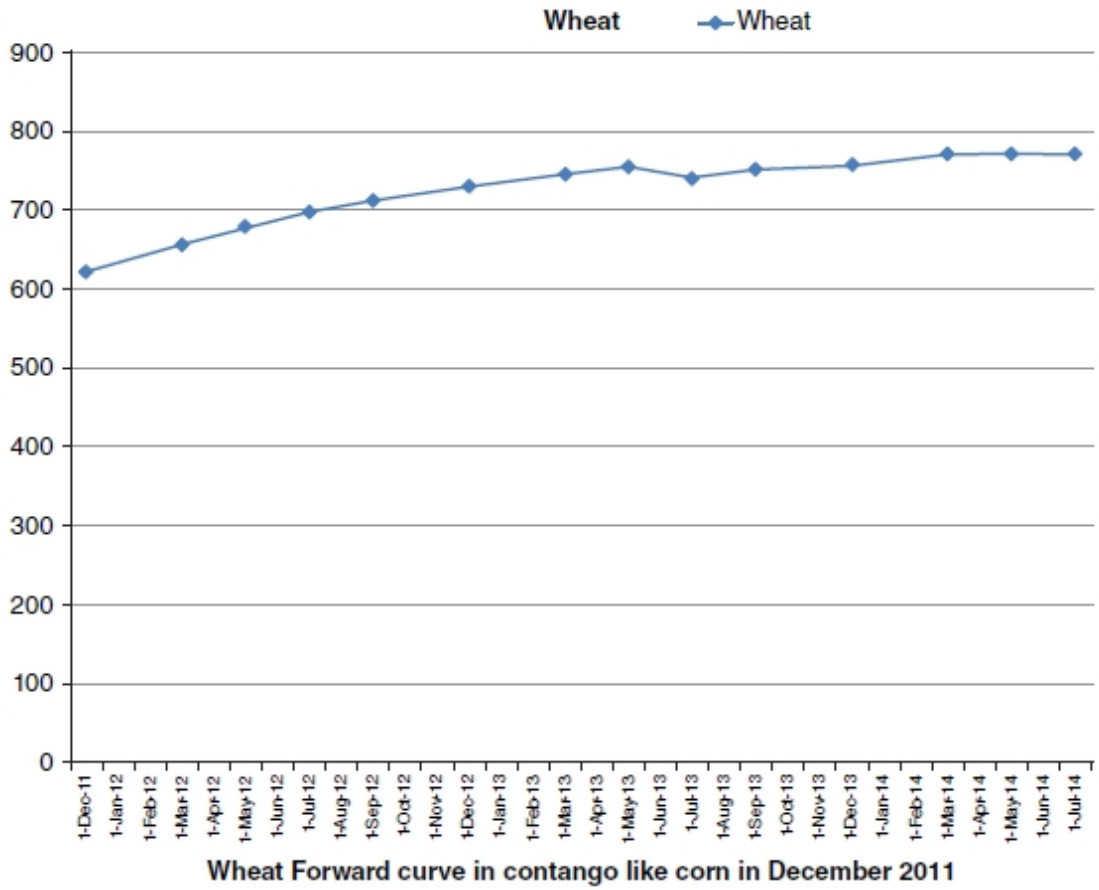


Figure: by buying the wheat spot using a loan, leasing part of a silo to store it, and selling a forward contract maturing, one could generate a sure profit at maturity after receiving the forward price and repaying the loan with accrued interest and the lease rate.

Example of a *backwardated* shape observed in corn

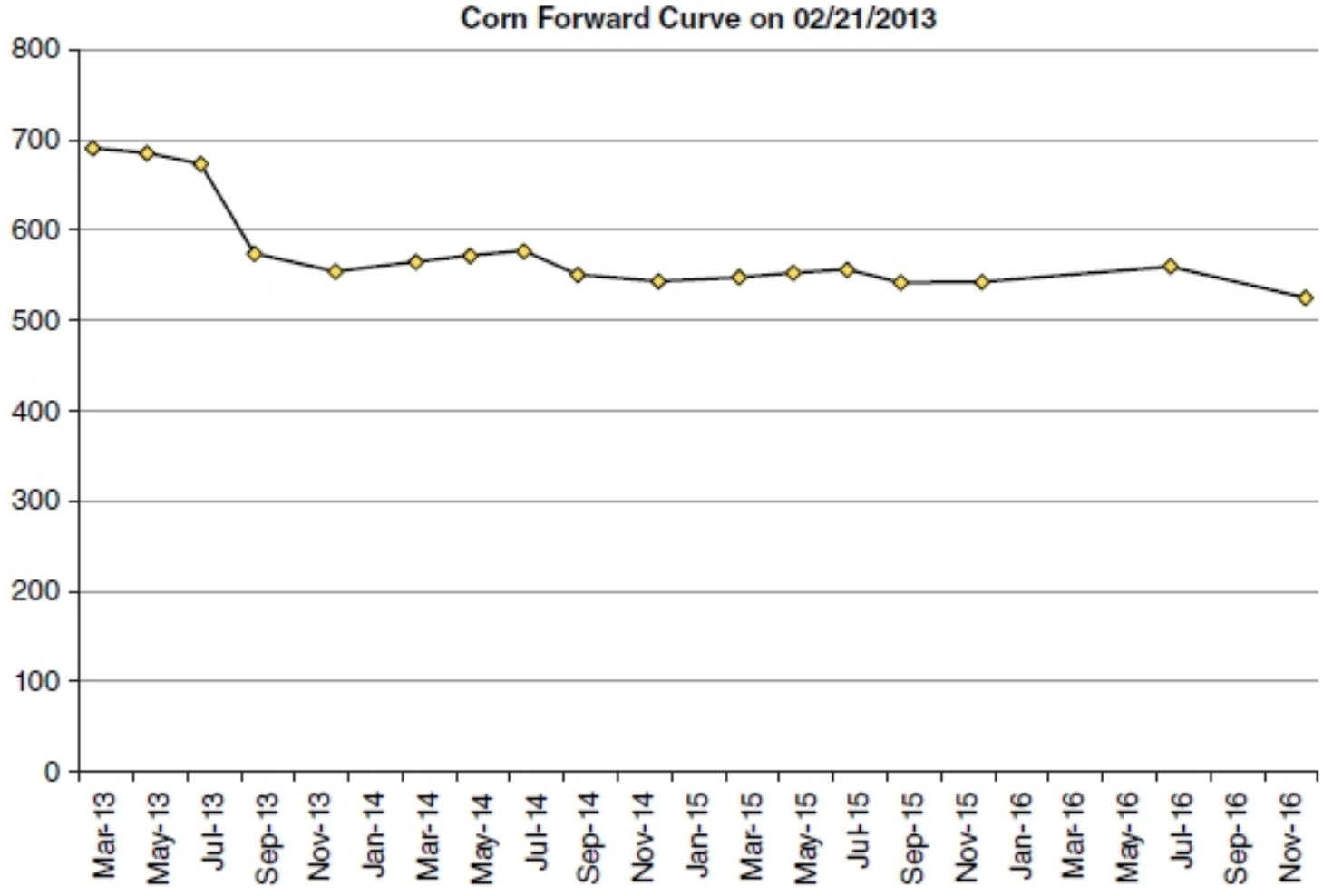


Figure: When there is a situation of scarcity, i.e., low inventory, spot prices will rise as purchasers bid whatever is necessary to secure supply

Contango forward curve for NYMEX crude oil

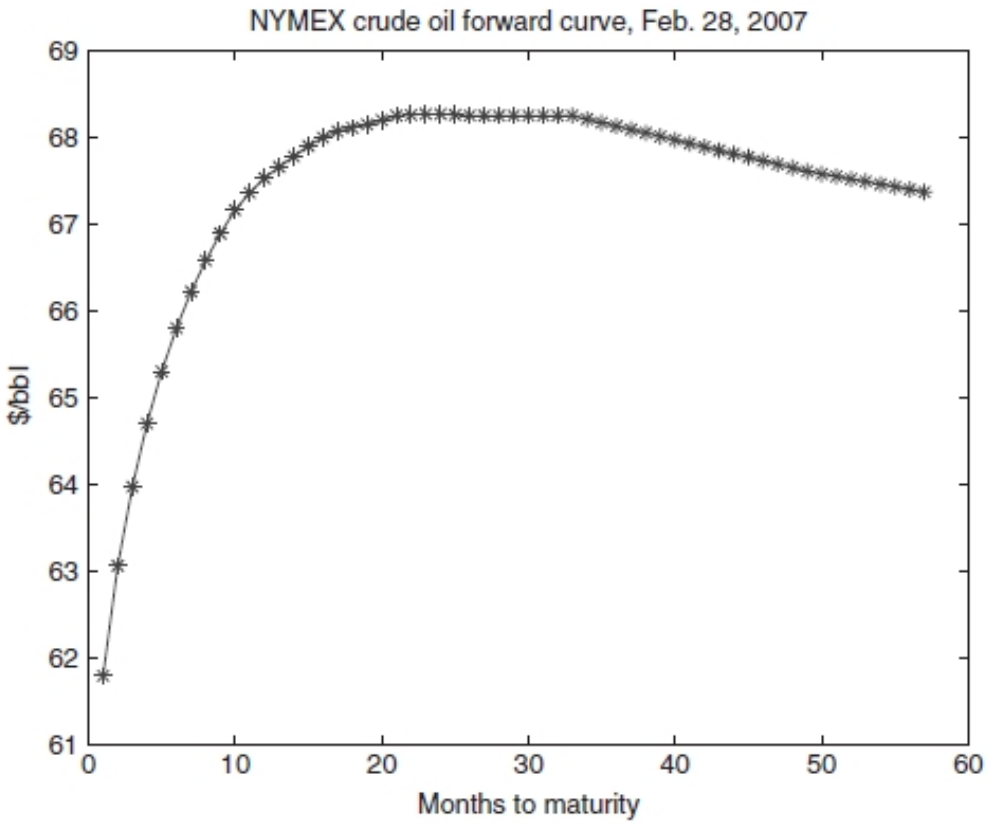


Figure: Whether the market is in *backwardation* or in *contango* depends on the current price as well as inventory levels, transportation and storage costs, supply and demand equilibria, strategic and political issues and many other factors. More generally, the shape of the forward curve summarizes perceptions of oil market participants about the current state and future developments of the global oil market, such as new fields discovery or new deep sea oil rig equipment.

Forward Curve Shapes: Summary

Contango vs. Backwardation

Contango: Forward prices increase with maturity ($F(t, T_2) > F(t, T_1)$ for $T_2 > T_1$)

- Occurs when: storage costs + interest > convenience yield
- Interpretation: ample supply, normal inventory levels
- Condition: $r + c > \delta$ where c = storage cost rate

Backwardation: Forward prices decrease with maturity

- Occurs when: convenience yield > storage costs + interest
- Interpretation: scarcity, tight market conditions
- Condition: $\delta > r + c$

Seasonal commodities

- For seasonal commodities such as natural gas, electricity, or agricultural commodities, futures prices are largely governed by seasonal demand (as is the case for energy) or supply (for agricultural commodities).
- For those commodities, transportation or storage capabilities are limited, so the excess demand or supply shortage cannot be absorbed by transporting a commodity from a different part of the world with excess supply, or storing at times of plentiful supply and using it whenever necessary.
- This results in a high seasonal premium on futures contracts maturing during periods of high demand (such as winter, in the case of natural gas) or low supply (such as before harvest, for agricultural commodities).

Pedagogical Takeaway

Seasonality introduces time-of-year effects in convenience yield: $\delta(t, T)$ depends on both time-to-maturity and calendar delivery date T.

Seasonality: NYMEX crude oil forward curve

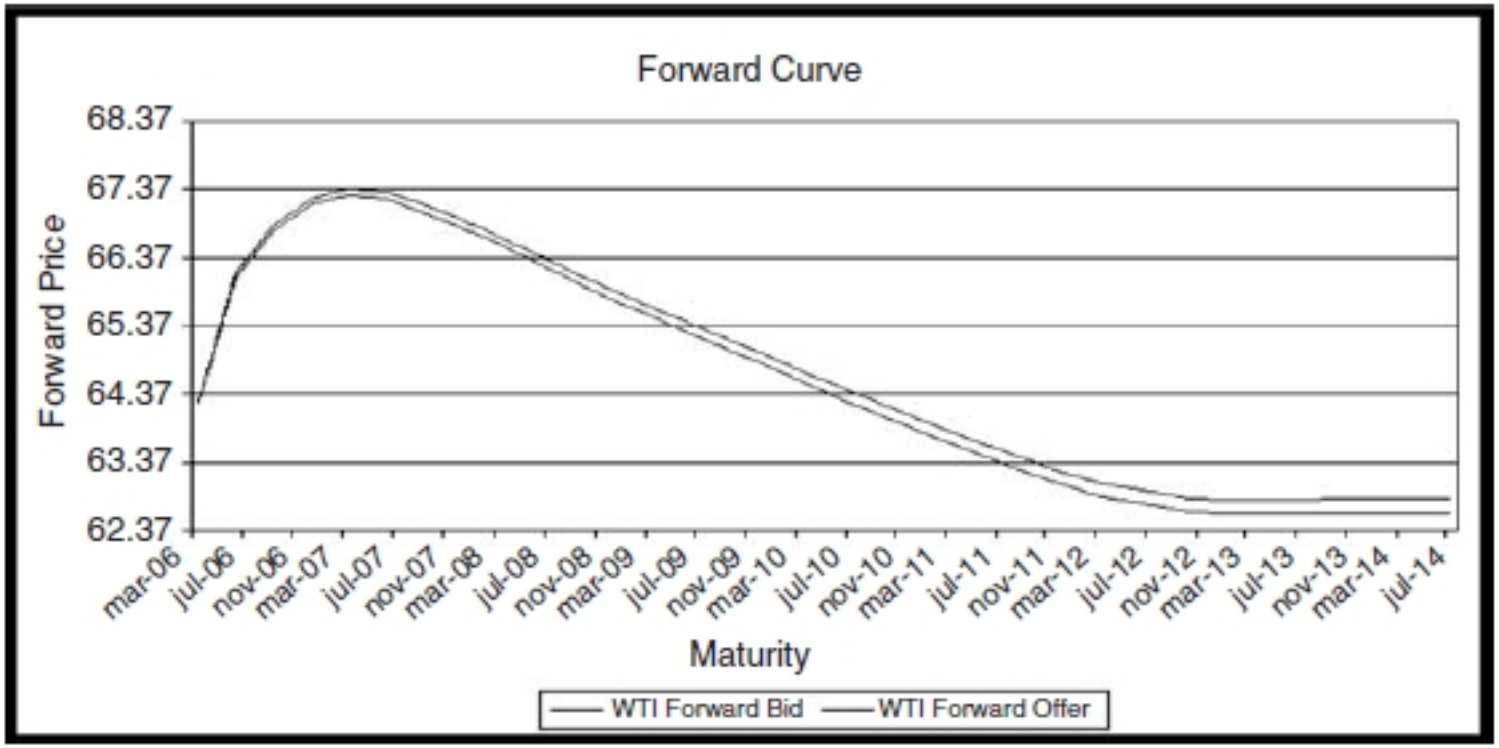


Figure: A hump-shaped forward curve was observed briefly, which was probably caused by a massive arrival of hedge funds and other new investors into oil futures markets and the short-dated futures rollover strategies they employed.

Seasonality: UK Natural gas forward curve

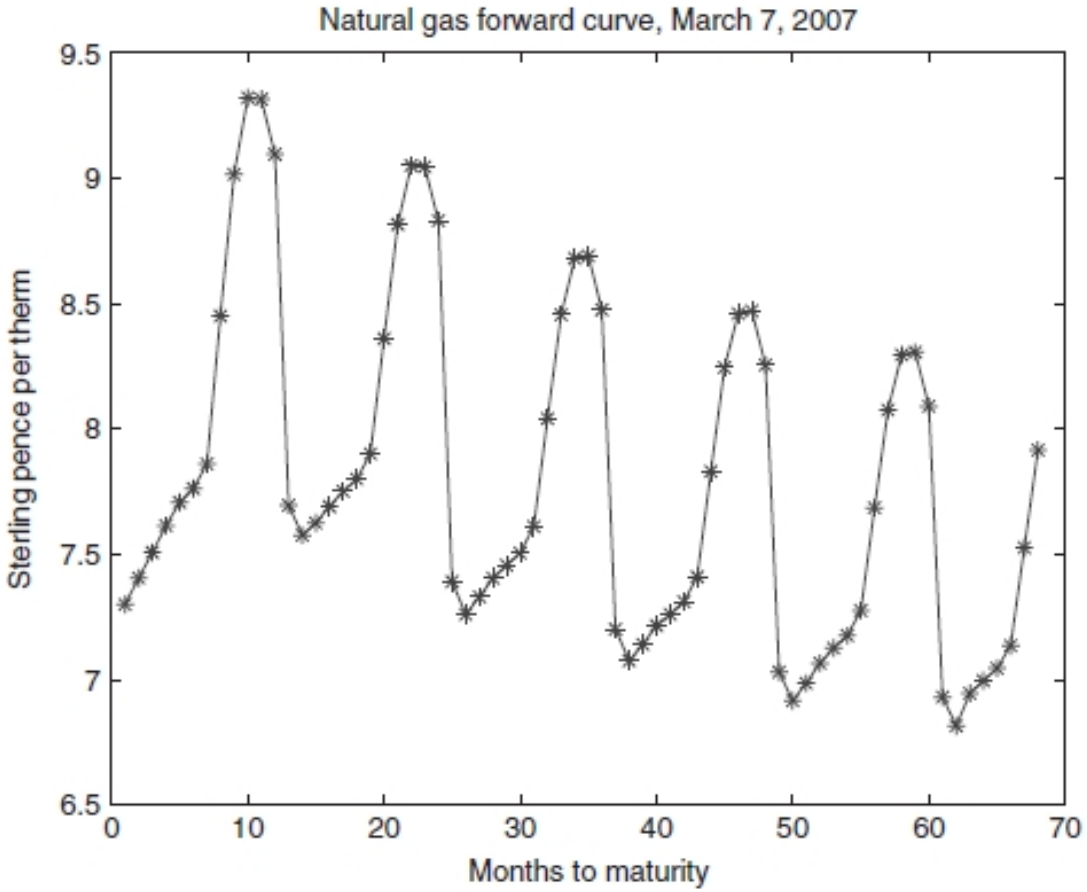


Figure: A forward curve for natural gas in the UK is shown. Futures maturing in the winter (the time of high demand for gas in the UK) are clearly at a premium compared to those maturing in summer.

Relation between global backwardation and deseasonalized inventory for US natural gas (1/2)

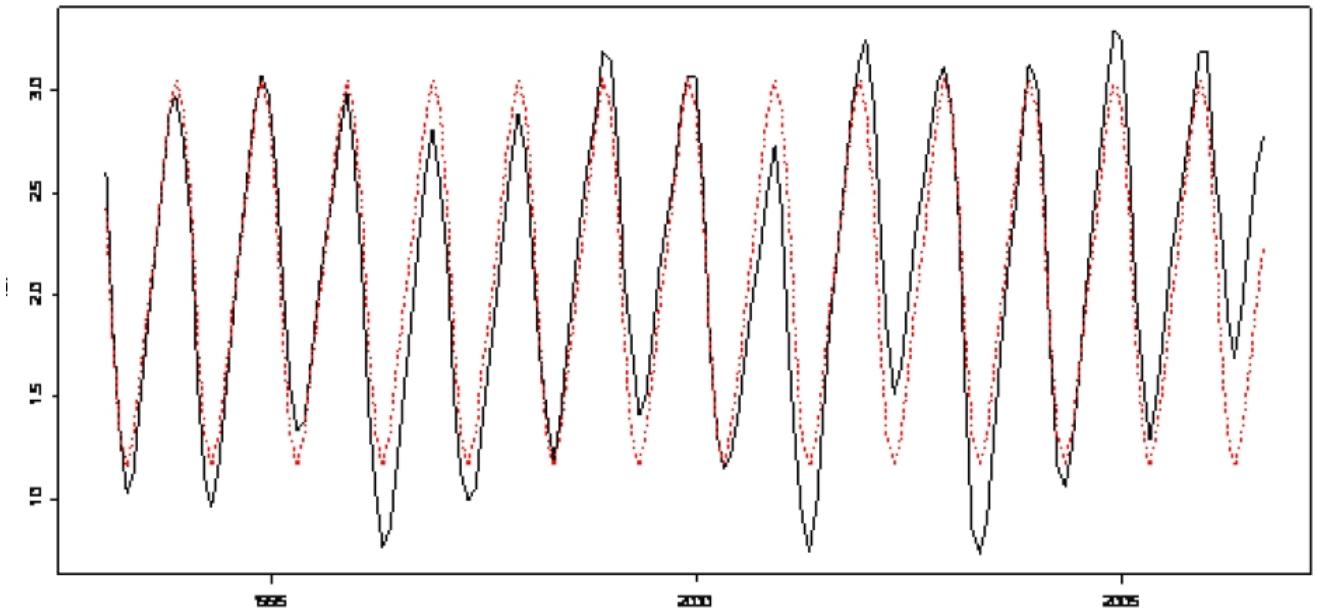


Figure: US Natural Gas inventory at the end of each month (in Trillion Cubic Feet) and trigonometric fit (dotted line) ; the difference between plain and dotted lines is the deviation to normal inventory.

Relation between global backwardation and deseasonalized inventory for US natural gas (2/2)

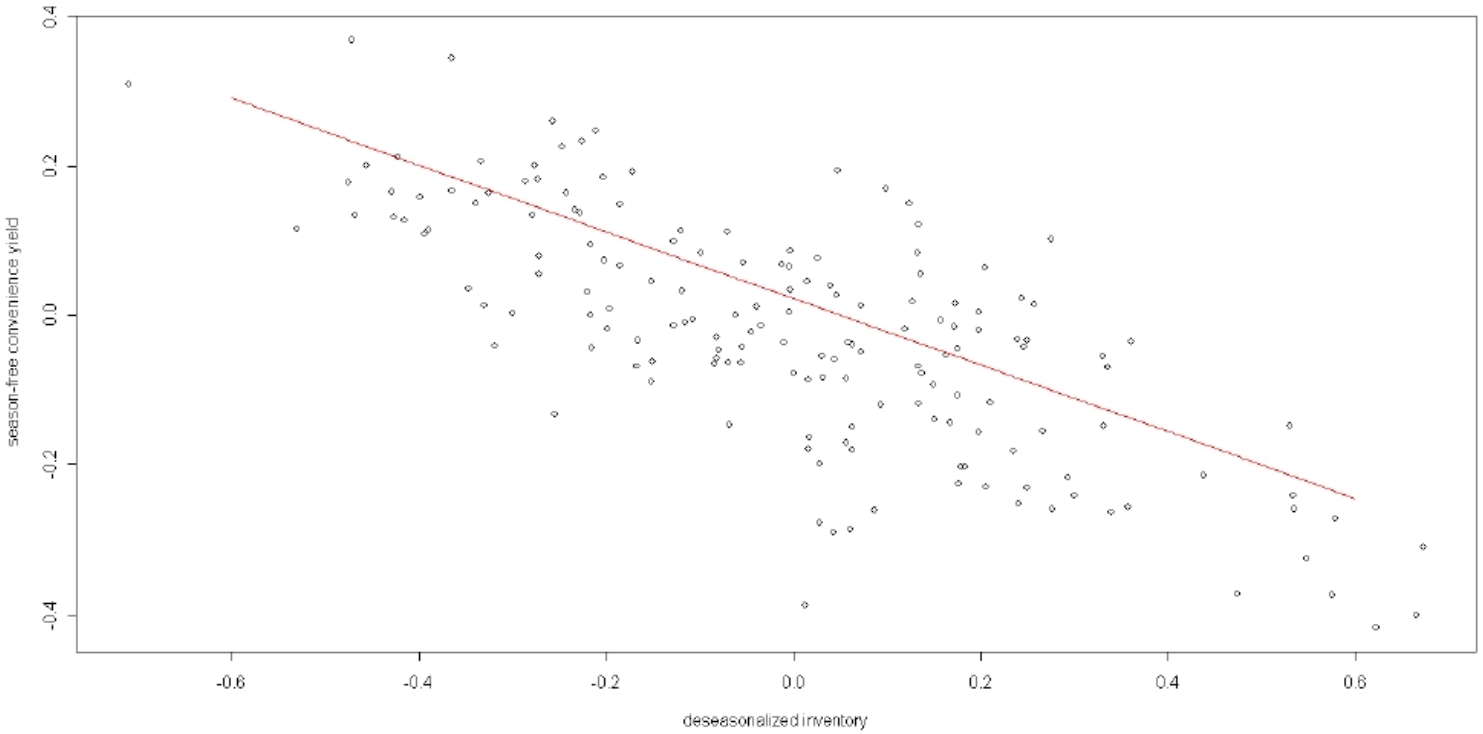


Figure: Natural Gas season-free convenience yield in terms of the deviation to normal inventory and best linear fit.

Gas forward curve moves in the US and UK markets (1/2)

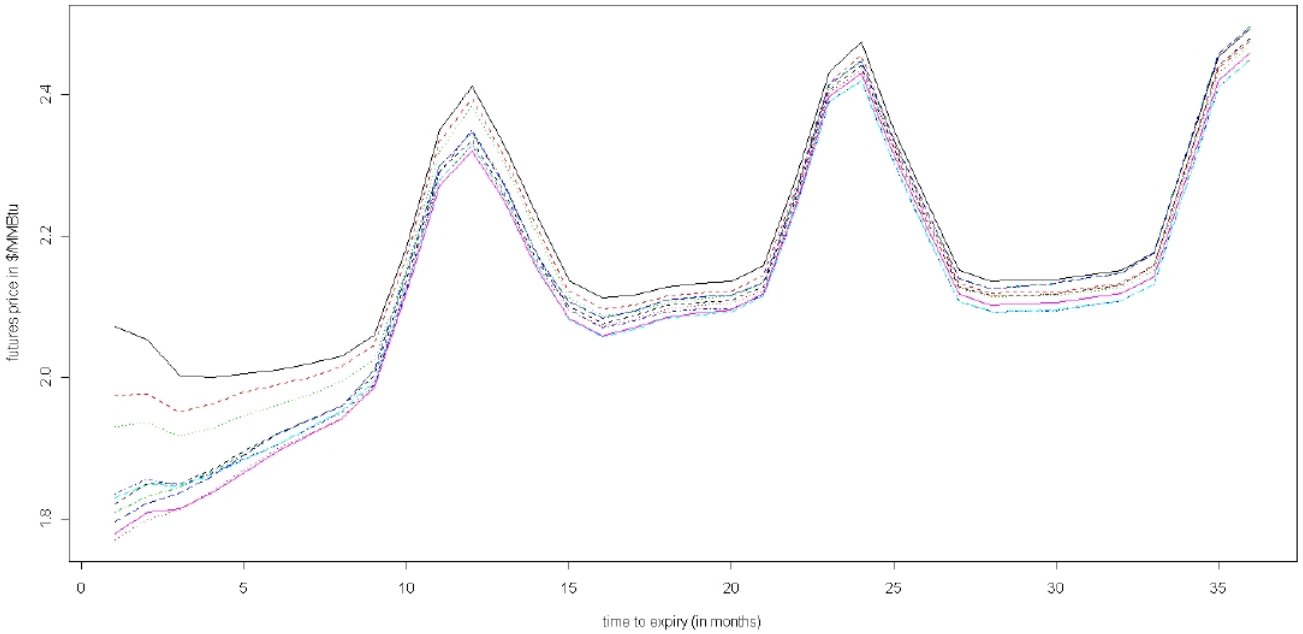


Figure: NYMEX gas forward curves.

Gas forward curve moves in the US and UK markets (2/2)

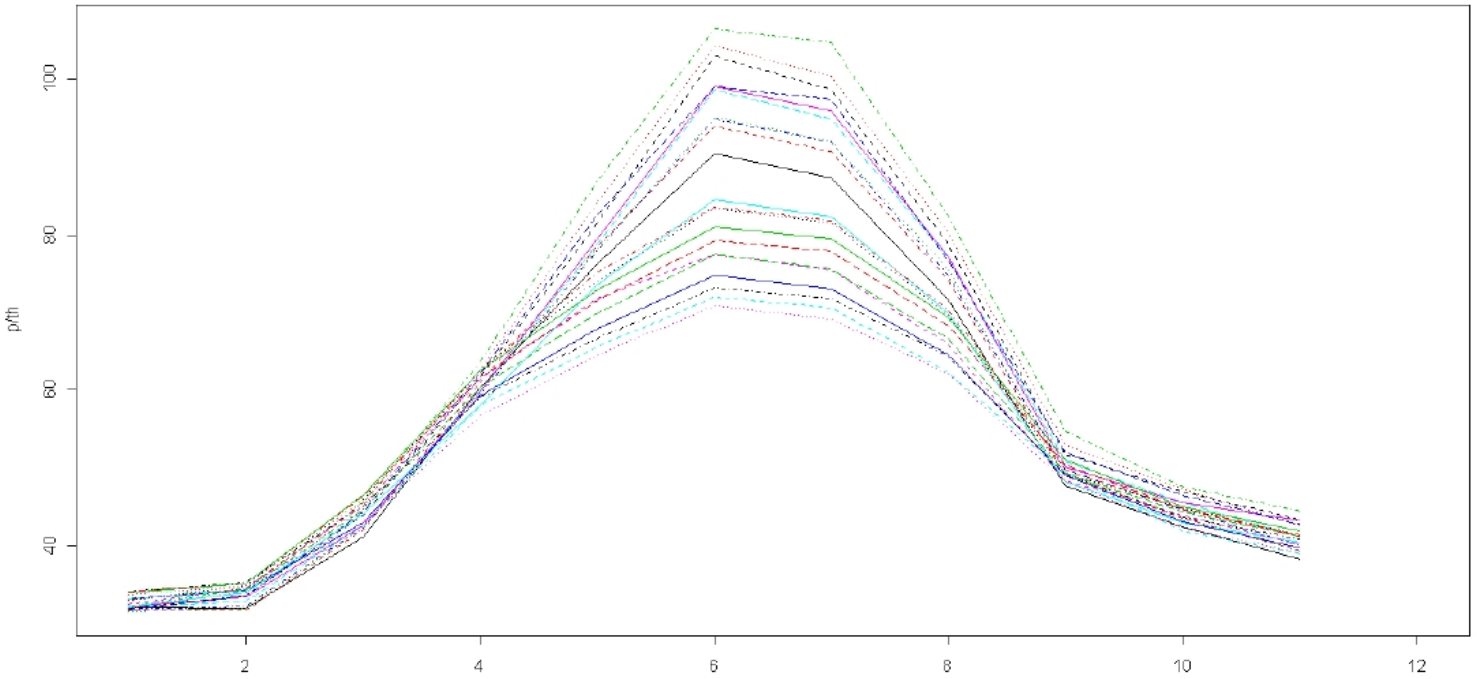


Figure: 'Hump'-shape in IPE gas forward curves. Apart from short-term and long-term shocks, *seasonal* shocks are also possible, affecting only the contracts for delivery in the winter.

Keynes' Insurance Perspective

Normal Backwardation (Keynes

Normal backwardation is the hypothesis that forward prices tend to lie *below* expected future spot prices because producers hedge price risk by selling forward and must offer a risk premium to induce speculators to take the opposite long position.

Mathematical Expression

$$F(t, T) = \mathbb{E}_t[S(T)] - \text{risk premium}$$

where the risk premium compensates speculators for bearing price risk.

Key Insights:

- Risk premium $= \mathbb{E}_t[S(T)] - F(t, T)$
- Forward price is a *biased predictor* of future spot
- Sign and magnitude vary with market structure and hedging pressure

Testing Normal Backwardation: Fama & French

Fama & French (1987) dual regression test

Test 1 — Forecasting power:

$$S_T - S_t = \alpha + \beta(S_t - F(t, T)) + \varepsilon$$

Examines whether basis predicts spot price changes.

Test 2 — Time-varying risk premium:

$$S_T - F(t, T) = \alpha + \beta(S_t - F(t, T)) + \varepsilon$$

Tests whether risk premium varies with basis (spot-forward spread).

Pedagogical Takeaway

Significant β in Test 2 supports time-varying premium linked to inventory levels. Empirical results are *mixed* across commodities—stronger for storables than non-storables.

Hedging Pressure: Definition and Measurement

Hedging Pressure

Hedging pressure measures the net demand by commercial hedgers to transfer price risk through forward or futures markets. It reflects the imbalance between producers' incentives to hedge by selling forward and consumers' incentives to hedge by buying forward, and it is the fundamental driver of the futures risk premium required by speculators.

Mathematical Formulation

$$HP_t = \frac{L_t^H - S_t^H}{L_t^H + S_t^H}$$

where:

- L_t^H = long positions held by commercial hedgers at time t
- S_t^H = short positions held by commercial hedgers at time t
- $HP_t \in [-1, 1]$

Alternative specification (basis regression):

$$\mathbb{E}_t[S_T] - F(t, T) = \alpha + \beta \cdot HP_t + \varepsilon_t$$

where $\beta > 0$ implies hedging pressure affects risk premium.

Hedging Pressure: Interpretation

Notation Summary

Key Variables:

- HP_t = hedging pressure index at time t
- L_t^H = long commercial hedger positions (open interest)
- S_t^H = short commercial hedger positions (open interest)
- L_t^S = long speculator positions
- S_t^S = short speculator positions
- $\mathbb{E}_t[S_T]$ = expected spot price at maturity T
- $F(t, T)$ = forward/futures price at t for delivery at T

Sign Interpretation:

- $HP_t > 0$: Net long hedging demand $\Rightarrow$ consumers dominate $\Rightarrow$ forward price bid up $\Rightarrow$ negative risk premium
- $HP_t < 0$: Net short hedging demand $\Rightarrow$ producers dominate $\Rightarrow$ forward price depressed $\Rightarrow$ positive risk premium
- $HP_t \approx 0$: Balanced hedging pressure $\Rightarrow$ minimal risk premium

Speculator Role and Market Equilibrium

Market Clearing Condition

In equilibrium, speculator positions offset hedger positions:

$$L_t^H - S_t^H = S_t^S - L_t^S$$

Speculators take the opposite side of hedger net positions.

Market Dynamics:

- **Speculators** = non-commercial traders providing liquidity
- Absorb risk from hedgers seeking price insurance
- Demand compensation (risk premium) for bearing directional exposure
- Their participation reduces risk premiums (improves market efficiency)
- In non-storable commodities (power, livestock), insurance view dominates storage theory

Pedagogical Takeaway

When producers dominate (net short hedgers), forward prices are depressed below expected spot, creating backwardation and positive returns for long speculators.

Empirical Evidence and Investment Implications

Data Source: CFTC Commitments of Traders (COT)

Weekly reports classify open interest into:

- **Commercial traders** (hedgers): producers, consumers, merchants
- **Non-commercial traders** (speculators): hedge funds, CTAs
- **Non-reportable** (small traders)

Used to compute HP_t for each commodity.

Investment implication (Gorton et al. 2007)

A strategy that goes *long* commodities with highest backwardation (strong net short hedging pressure) and *short* those in strongest contango (strong net long hedging pressure) tends to outperform long-only portfolios.

Typical findings:

- Hedging pressure explains 5-15% of risk premium variation
- Effect stronger in agricultural commodities than energy
- Time-varying—increases during volatile periods

Samuelson Effect: Definition

Samuelson Effect (1965)

The empirical observation that the **volatility of futures prices decreases with time to maturity**. Near-term contracts are more volatile than distant contracts.

Economic Rationale:

- Short-term shocks (weather, inventory news, supply disruptions) primarily affect nearby contracts
- Long-dated contracts reflect fundamental supply-demand balance
- Production can adjust over longer horizons, dampening price volatility
- Market participants have more information about near-term conditions

Pedagogical Takeaway

Samuelson effect is strongest for storable commodities with active spot markets and weakest for commodities with long production lags or limited storability.

Term structure of futures price volatility for Crude Oil and Natural Gas in the NYMEX

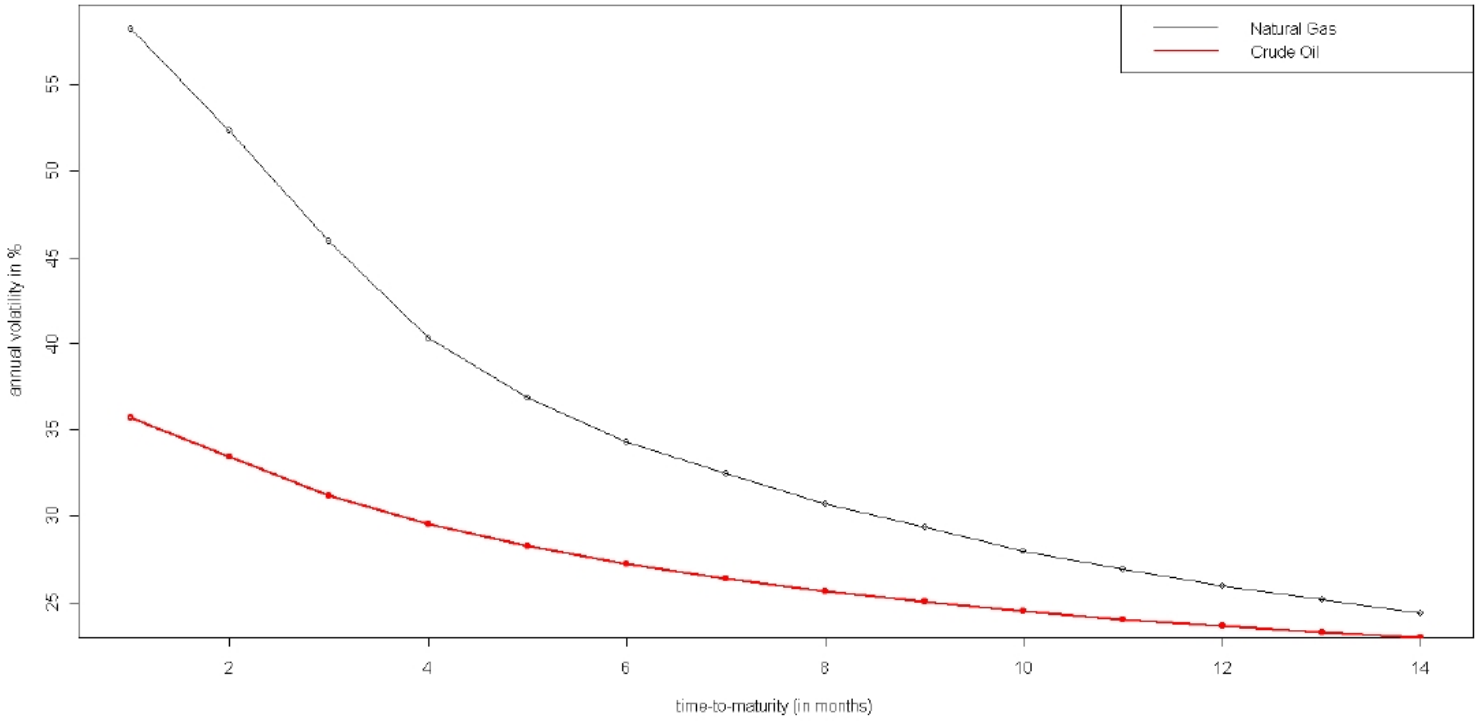


Figure: Representing the futures price volatility as a function of time to maturity, we clearly observe that the volatility of the futures prices decreases with time-to-maturity, a phenomenon referred to as the "Samuelson effect".

Why Does the Samuelson Effect Arise?

Two types of shocks drive forward curve movements:

1. **Long-term shocks**: affect all maturities similarly
   - Technology changes
   - Major reserve discoveries
   - Long-term demand shifts
   - Regulatory regime changes
2. **Short-term shocks**: affect only front months
   - Pipeline congestion
   - Weather anomalies
   - Temporary inventory swings
   - Short-lived supply disruptions

Link to theory of storage

During scarcity, convenience yield is high, spot volatility rises sharply, but long-dated futures remain relatively stable as long-term fundamentals unchanged.

HJM Model for Commodity Forward Curves

Heath-Jarrow-Morton Framework

A general arbitrage-free framework for modeling the dynamics of an entire forward curve, adapted from interest rate theory to commodities.

General dynamics under statistical measure $\mathbb{P}$

$$\frac{dF(t,T)}{F(t,T)} = \alpha(t,T)\,dt + \sum_{i=1}^{K} \sigma_i(t,T)\,dW_t^i$$

where

- $\alpha(t,T)$ = drift term structure (risk premium)
- $\sigma_i(t,T)$ = volatility term structure for factor i
- dW_t^i = correlated Brownian shocks (can be non-Gaussian)
- K = number of risk factors

Drift Term $\alpha(t,T)$:
The drift term $\alpha(t,T)$ captures the **risk premium** associated with holding the commodity forward contract. It represents the compensation required by market participants for bearing the risk of price fluctuations in the forward market.

HJM Framework: Risk-Neutral Dynamics

No-Arbitrage Condition (HJM Drift Restriction)

Under the **risk-neutral measure** $\mathbb{Q}$, forward prices must satisfy:

$$\alpha(t, T) = \sum_{i=1}^{K} \lambda_i(t)\, \sigma_i(t, T)$$

This ensures no arbitrage opportunities exist.

- $\alpha(t, T)$: Drift term under $\mathbb{P}$ (real-world measure).
- $\lambda_i(t)$: **Market price of risk** for factor i, linking $\mathbb{P}$ and $\mathbb{Q}$.
- $\sigma_i(t, T)$: Volatility term structure for factor i.

Notation Summary

Key Notation:

- $\mathbb{P}$: Real-world (statistical) probability measure.
- $\mathbb{Q}$: Risk-neutral (pricing) probability measure.
- $\widetilde{W}_t^i$: Brownian motion under $\mathbb{Q}$.

HJM Framework: Risk-Neutral Dynamics (cont.)

Forward Price Dynamics Under $\mathbb{Q}$

Under the risk-neutral measure $\mathbb{Q}$, the forward price dynamics simplify to:

$$\frac{dF(t,T)}{F(t,T)} = \sum_{i=1}^{K} \sigma_i(t,T)\, d\widetilde{W}_t^i$$

Martingale Property

Forward prices are **martingales** under $\mathbb{Q}$:

$$\mathbb{E}_t^{\mathbb{Q}}[F(t_1,T)] = F(t_0,T) \quad \forall t_0 \leq t_1 \leq T$$

This means: **No-arbitrage implies that forward prices are unbiased predictors of future forward prices** (but not necessarily of spot prices!).

Pedagogical Takeaway

Key Takeaway: The no-arbitrage constraint removes the drift under $\mathbb{Q}$, ensuring forward prices are fair predictors of future forward prices.

Why Do We Need $\mathbb{P}$ and $\mathbb{Q}$?

Real World ($\mathbb{P}$) vs. Risk-Neutral World ($\mathbb{Q}$)

- In the **real world ($\mathbb{P}$)**, forward prices include a **risk premium** ($\alpha(t, T)$) because investors demand compensation for bearing risk.
- For **pricing derivatives**, we need a world without risk premiums to ensure **no-arbitrage**. This is the **risk-neutral measure ($\mathbb{Q}$)**.

Role of the HJM Framework

The HJM framework models the **entire forward curve** while ensuring:

- Forward prices are **consistent with no-arbitrage** under $\mathbb{Q}$.
- The drift term $\alpha(t, T)$ is replaced by $\sum_{i=1}^{K} \lambda_i(t)\, \sigma_i(t, T)$ to account for the market price of risk.

Pedagogical Takeaway

Key Idea: $\mathbb{P}$ describes reality, while $\mathbb{Q}$ ensures fair, arbitrage-free pricing of derivatives.

Principal Component Analysis (PCA) – Idea

Principal Component Analysis (PCA)

A statistical technique that extracts uncorrelated risk factors (principal components) from historical forward curve movements by diagonalizing the covariance matrix of returns.

Implementation Steps:

1. Compute daily log-returns for each maturity:

$$r_t^i = \frac{F(t, T_i) - F(t - \Delta t, T_i)}{F(t - \Delta t, T_i)} \approx \ln\left(\frac{F(t, T_i)}{F(t - \Delta t, T_i)}\right)$$

2. Build covariance matrix Σ of returns across N maturities.
3. Diagonalize: $\Sigma = PDP^\top$ where P = eigenvector matrix, D = diagonal eigenvalue matrix.
4. Eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$ represent variance explained by each PC.
5. First few eigenvectors describe main deformation patterns.

PCA Results: Typical Pattern

Typical result for commodities

- **PC1 (level/parallel shift)**: all maturities move together
  - Explains $\sim 90\%$ of variance
  - Uniform sign across all maturities
- **PC2 (slope/tilt)**: near vs. far maturities move oppositely
  - Explains $\sim 5\%$ of variance
  - Sign changes from short to long end
- **PC3 (curvature/twist)**: middle vs. ends move oppositely
  - Explains $\sim 1\%$ of variance
  - Two sign changes along maturity spectrum

Total: Three factors capture $> 95\%$ of variance for crude oil, natural gas

Pedagogical Takeaway

PCA dimensionality reduction: infinite-dimensional forward curve dynamics compressed to 2-3 factors with minimal information loss.

PCA Loadings for Crude Oil and Diesel

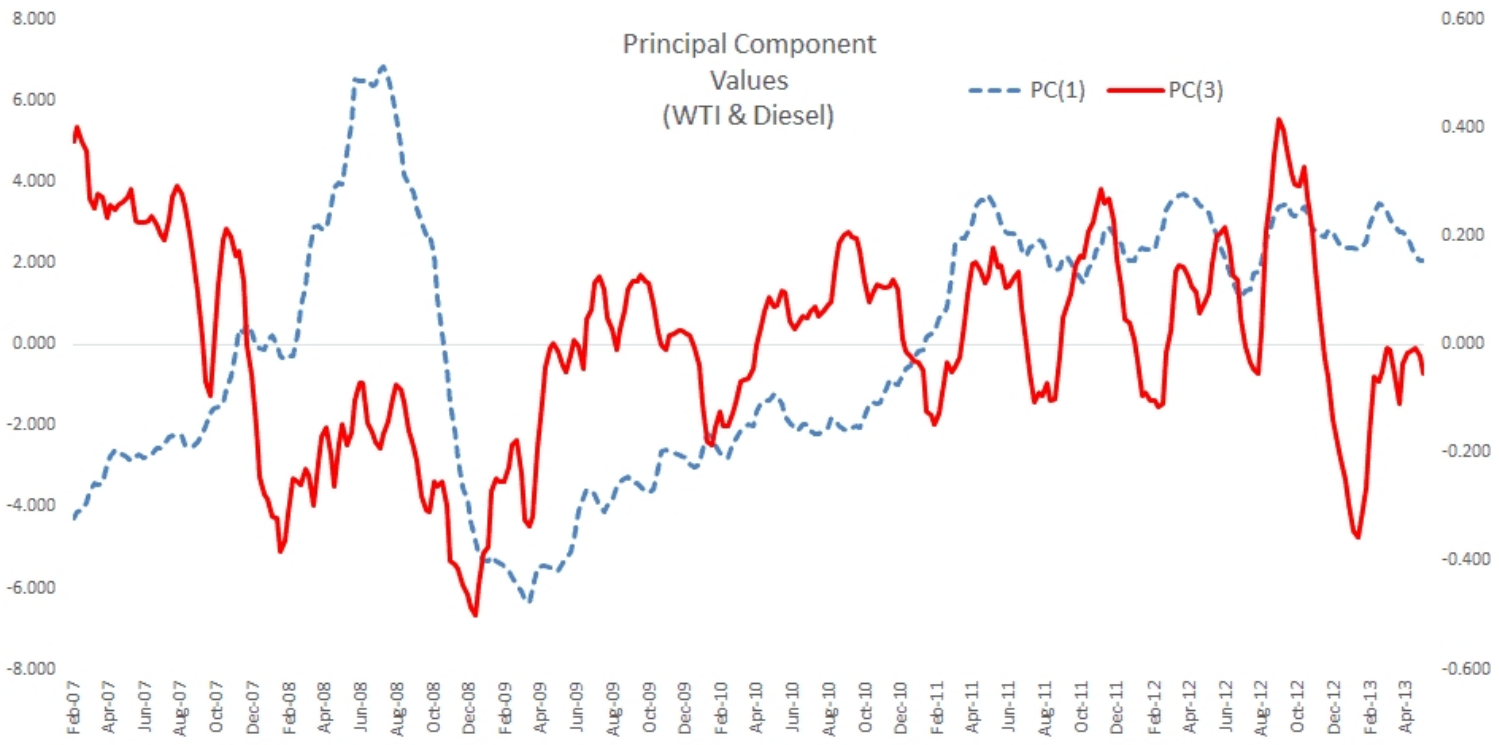


Figure: Example of principal component loadings.

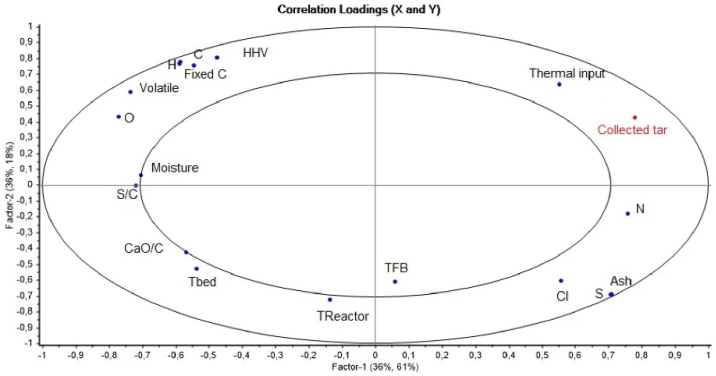
Interpretation:

- PC1: Positive loadings $\Rightarrow$ all contracts rise/fall together
- PC2: Front/back opposite signs $\Rightarrow$ slope steepens/flattens
- PC3: Middle different from ends $\Rightarrow$ curve bends

PCA example #2

a)

b)



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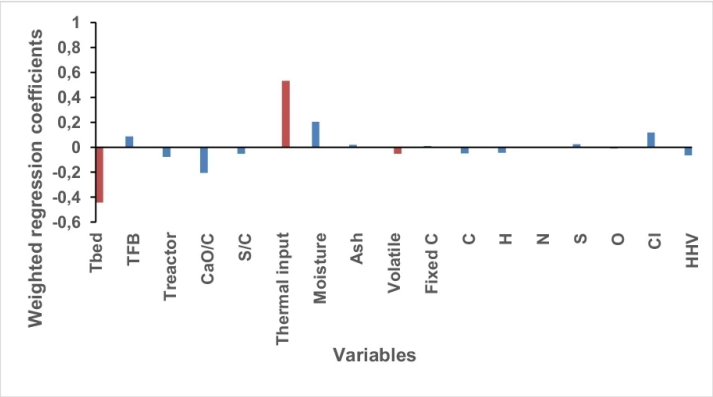
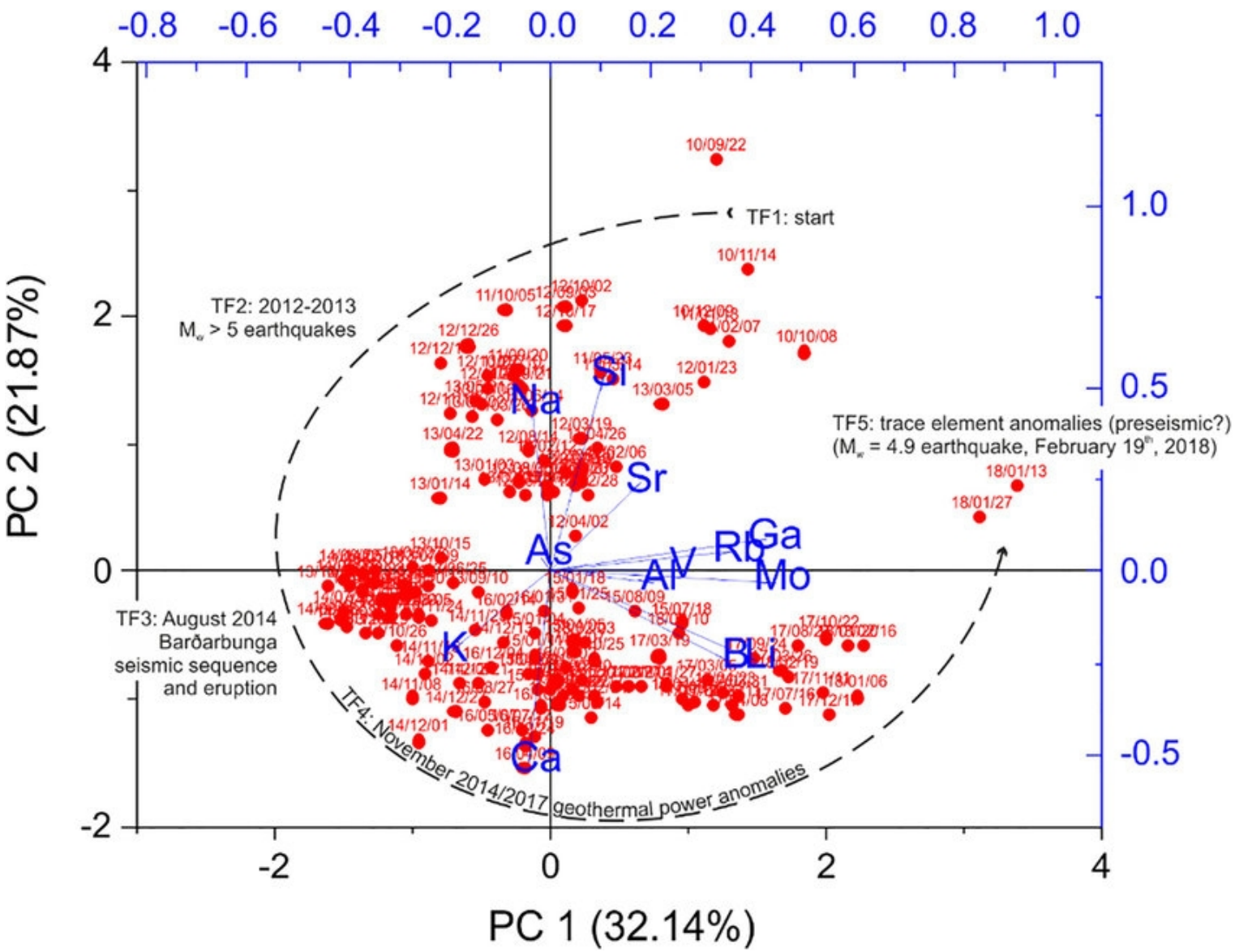


PCA example #3



PCA: Practical Considerations

Implementation Notes:

- **Deseasonalize first**: For seasonal commodities (gas, ags), remove deterministic seasonal component before PCA
- **Rolling window**: Re-estimate PCs periodically (monthly/quarterly) as market structure evolves
- **Data quality**: Requires liquid, reliable quotes across maturity spectrum
- **Model-free**: PCA makes no assumptions about functional form—purely data-driven

Notation Summary

PCA Reconstruction:

$$r_t = \sum_{i=1}^{K} w_i \cdot PC_i \cdot \text{Loading}_i$$

where w_i = PC score (time series), Loading_i = eigenvector (maturity profile)

Two-Factor Exponential Model

Parametric Two-Factor Model

A parsimonious model that separates forward curve dynamics into mean-reverting short-term shocks and persistent long-term shocks with exponential decay.

Parametric volatility specification

$$\frac{dF(t,T)}{F(t,T)} = e^{-k(T-t)}\, dX_t + dY_t$$

where

- dX_t = short-term shock (mean-reverting, high volatility)
  - Decays exponentially: $e^{-k(T-t)} \to 0$ as $T \to \infty$
- dY_t = long-term shock (random walk, lower volatility)
  - Affects all maturities equally
- k = mean-reversion speed of short-term factor (units: time^{-1})
  - Larger $k \Rightarrow$ faster reversion $\Rightarrow$ steeper Samuelson effect

Two-Factor Model: Economic Interpretation

Shock Dynamics

Short-term component X_t:

$$dX_t = [\sigma_X \lambda_X - kX_t]\, dt + \sigma_X\, dW_t^X$$

- Mean-reverts to risk premium $\sigma_X \lambda_X / k$
- Represents temporary supply-demand imbalances
- Examples: weather, inventory surprises, congestion

Long-term component Y_t:

$$dY_t = \sigma_Y \lambda_Y\, dt + \sigma_Y\, dW_t^Y$$

- Persistent drift (can be set to zero under $\mathbb{Q}$)
- Represents fundamental structural changes
- Examples: technology shifts, reserve discoveries, regime changes

Correlation structure:

$$\mathrm{Corr}(dX_t, dY_t) = \rho \quad \text{(typically 0.4–0.5 for energy)}$$

Two-Factor Model: Key Properties

Notation Summary

- k = mean-reversion speed (dimension: time^{-1})
- $1/k$ = characteristic time scale of short-term shocks
- σ_X = volatility of short-term factor
- σ_Y = volatility of long-term factor
- ρ = correlation between X and Y factors

Samuelson Effect Captured:

$$\sigma_{F(t,T)} = \sqrt{\sigma_X^2 e^{-2k(T-t)} + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y e^{-k(T-t)}}$$

Volatility decreases monotonically with $(T - t)$ as exponential terms vanish.

Pedagogical Takeaway

Model advantages: (1) Analytical tractability, (2) Captures Samuelson effect, (3) Only 5 parameters. **Limitation:** May underfit cross-maturity correlations compared to 3-factor PCA models.

Calibration and Fit

Calibration steps:

1. **Choose** k: Minimize root mean squared error (RMSE) between model and observed returns across all maturities.

$$\text{RMSE} = \sqrt{\frac{1}{N \times M} \sum_{t=1}^{N} \sum_{i=1}^{M} \left[\left(\frac{\Delta F(t, T_i)}{F(t, T_i)} \right)^{\text{obs}} - \left(\frac{\Delta F(t, T_i)}{F(t, T_i)} \right)^{\text{model}} \right]^2}$$

2. **Extract shocks**: From two liquid maturities (e.g., 1-month and 14-month).

$$dY_t = \left(\frac{\Delta F(t, T_{14})}{F(t, T_{14})} \right)^{\text{obs}}, \quad dX_t = e^{k(T_1 - t)} \left[\left(\frac{\Delta F(t, T_1)}{F(t, T_1)} \right)^{\text{obs}} - dY_t \right]$$

3. **Estimate statistics**: Compute $\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho$ from extracted shock time series.

Calibration Results

Parameter	Crude oil	Natural gas
k (month^{-1})	2.6	3.21
Short-term char. time (months)	4.6	3.8
Unexplained variance (RMSE %)	1.3	5.5

Table: Calibration results for two-factor exponential model (Ohana, 2006).

Interpretation:

- Gas has faster mean-reversion (shorter memory): 3.8 vs. 4.6 months
- Gas fit is poorer (5.5% unexplained) due to stronger seasonal/twist components
- Oil model captures 98.7% of volatility structure

Pedagogical Takeaway

Higher unexplained variance for gas suggests need for 3-factor model or explicit seasonality for pricing applications requiring precise cross-maturity correlation fits.

Fit of the term structure of futures price volatility (1/2)

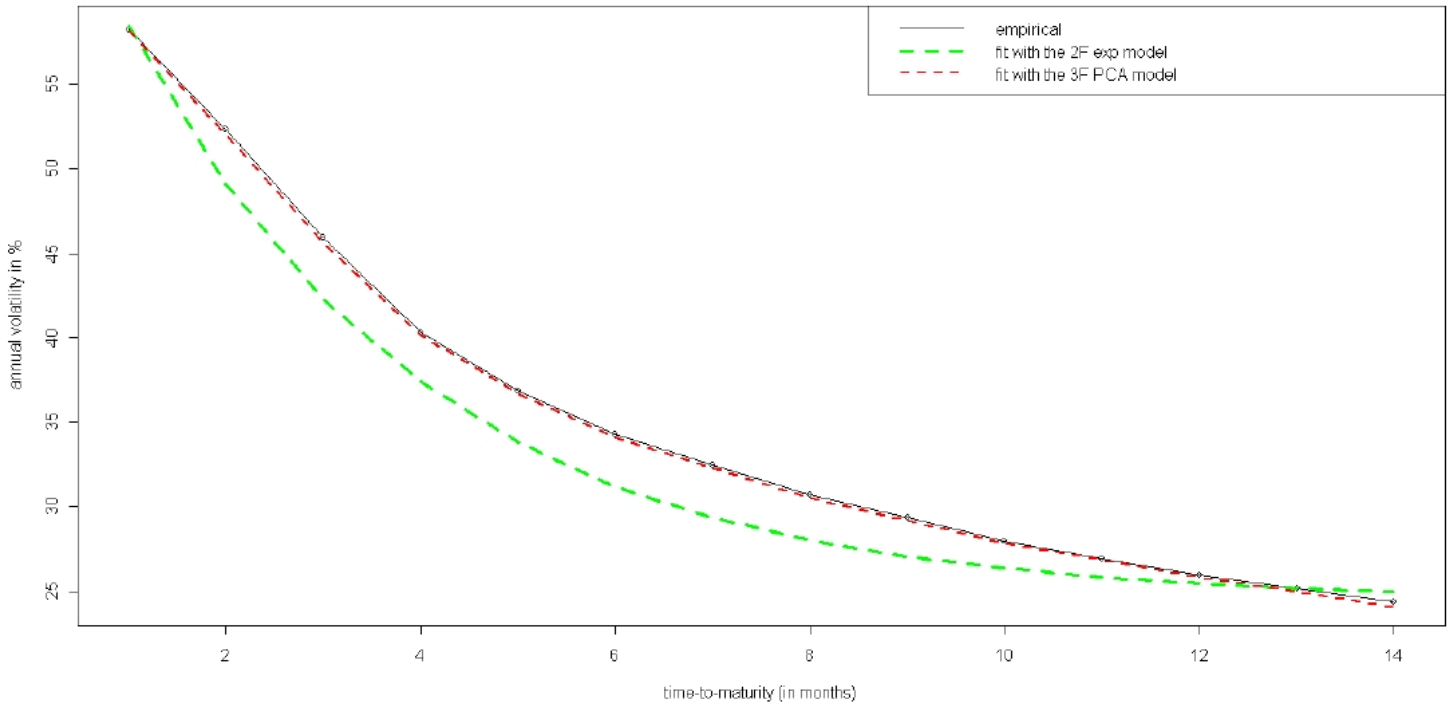


Figure: Natural Gas: the fit is correct with the two-factor exponential model, and almost perfect with the the three-factor model calibrated by PCA.

Fit of the term structure of futures price volatility (2/2)

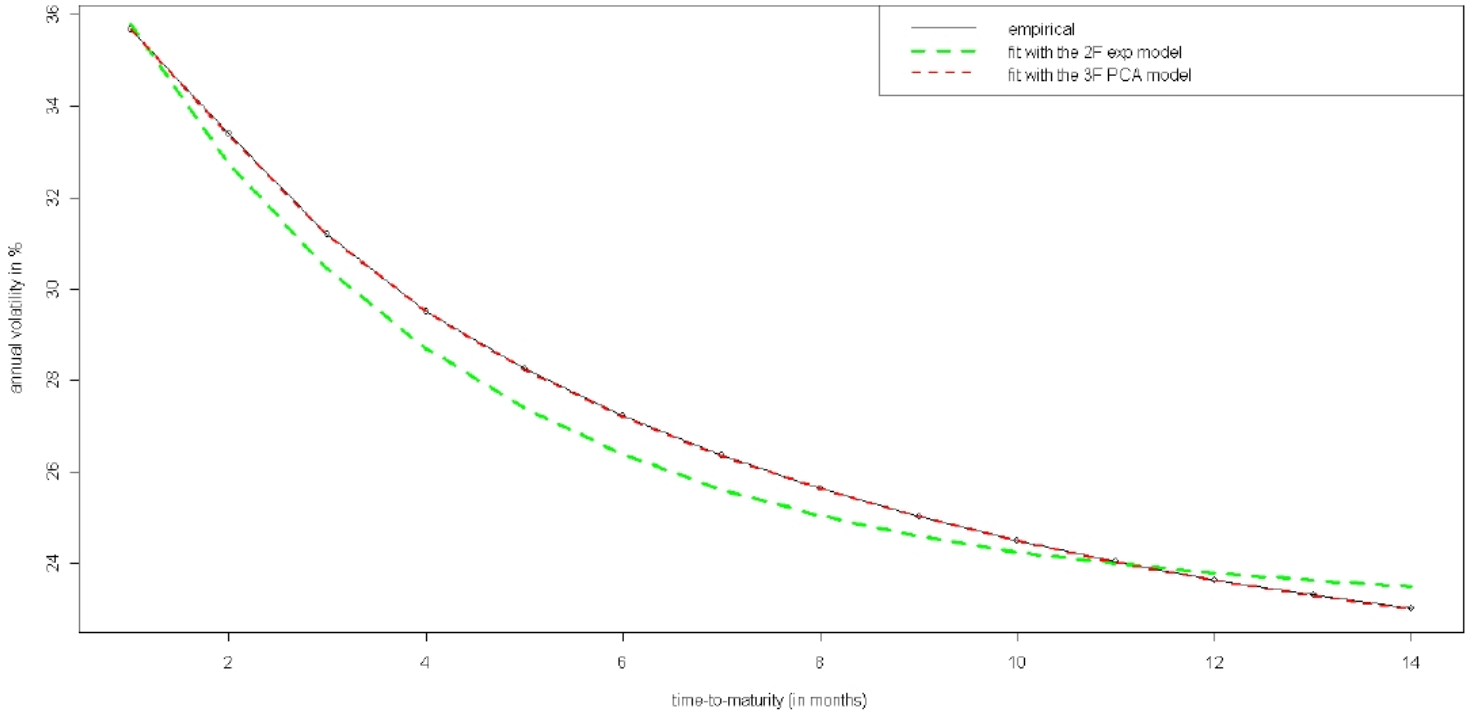


Figure: Crude Oil: the factor models have to be employed with great care in applications which require a very precise fit of cross-maturity correlations (e.g. measuring the risk of long/short positions or pricing calendar spread options).

Key Takeaways

Theory of Storage

Links spot and forward prices via **convenience yield** $\delta(t, T)$, reflecting physical market tightness (inventory levels).

Keynes' Normal Backwardation

Introduces **risk premium** as compensation for bearing price risk.
$\Rightarrow$ Forward prices may be **biased predictors** of future spot prices.

Samuelson Effect

Forward volatility $\sigma_F(T-t)$ **decreases** with time-to-maturity due to mean-reversion of short-term shocks.

HJM Framework

Provides **arbitrage-free dynamics** for the entire forward curve with drift restriction:

$$\alpha(t, T) = \sum \lambda_i(t)\sigma_i(t, T)$$

PCA Approach

Extracts dominant deformation modes:

- Level ($\sim$90%)
- Slope ($\sim$5%)
- Curvature ($\sim$1%)

$\Rightarrow$ **Model-free** approach.

Two-Factor Exponential Models

Separates:

- **Short-term shocks** (mean-reverting, decay $e^{-k(T-t)}$)
- **Long-term shocks** (persistent)

$\Rightarrow$ Parsimoniously captures the Samuelson effect.

Practical Implementation: From Theory to Action

Model Selection Guide

Choose based on application:

Simple Instruments: **2-Factor Exponential** (fast, analytical)

Spread Options: **3-Factor PCA** (captures cross-maturity correlations)

Seasonal Commodities: Add **deterministic seasonality**

Complex Exotics: **Full HJM** + stochastic volatility/jumps

Core Engineering Principle

Balance complexity and stability.
More parameters $\neq$ better performance
if estimates are unstable.

Rule of Thumb: Start simple. Add complexity **only** when data justifies it.

Key Takeaway:
Model selection is a trade-off.
Choose the simplest model that captures the risk factors you need to manage.

Questions? (or specific applications to discuss?)